

External Input

Estimated BOM cost: \$0.03

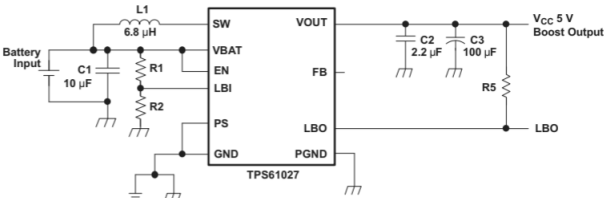


Alt Power Input

File: ExternalInput.kicad_sch

Boost Converters

Estimated BOM cost: \$0.60



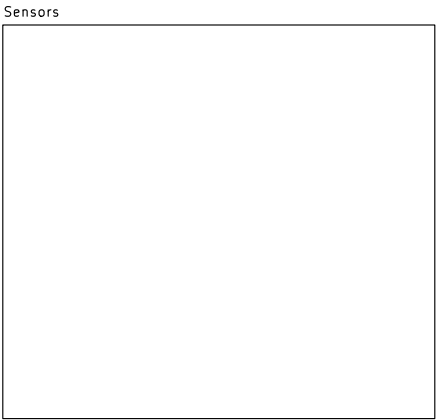
List of Components:
U1 = TP561027DRC
L1 = EPCOS B62462-G4682
C1, C2 = X7R,X5R Ceramic
C3 = Low ESR Tantalum

Figure 24. Power Supply Solution for Maximum Output Power Operating From a Single Alkaline Cell

Power Conversion

File: Boost.kicad_sch

Sensors



File: Sensors.kicad_sch

USB Input

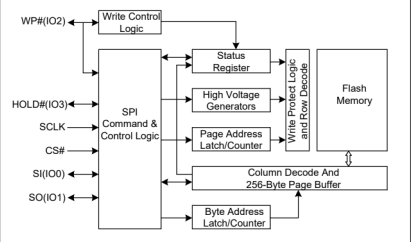
Estimated BOM cost: \$1.00
(\$0.02 without serial to usb)



USB Input
(Power/serial)

File: USB_Input.kicad_sch

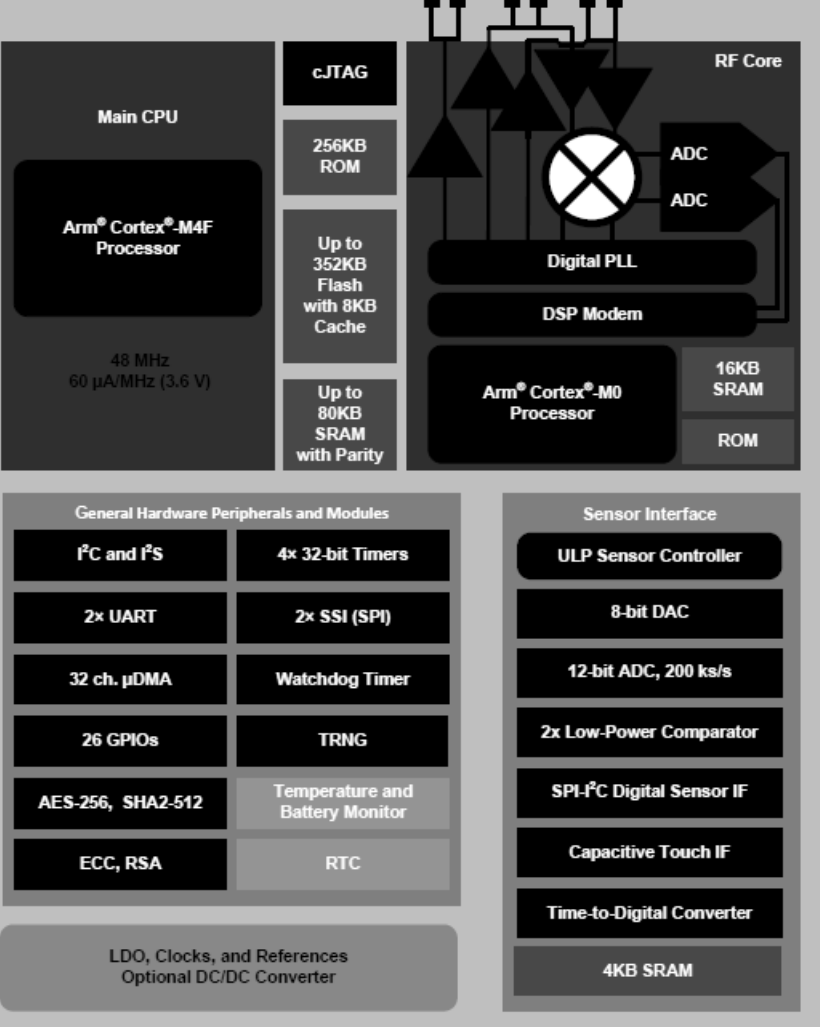
SPI FLASH



File: SPI_FLASH.kicad_sch

RF

Estimated BOM cost: \$1.27

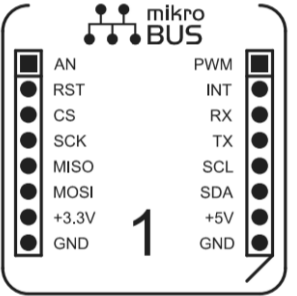
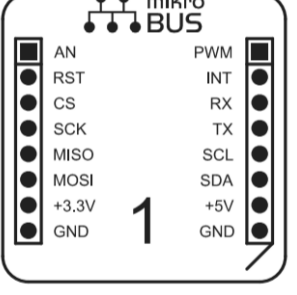


RF and Processor

File: RF.kicad_sch

MikroBus Click

Estimated BOM cost: \$0.15

MikroBus Headers

File: MikroBusClick.kicad_sch

Sheet5D099F01

Estimated BOM cost: \$0.03



Indicators

File: Indicators.kicad_sch

H1 H2 H3



GND

User Input/Output

Estimated BOM cost: \$0.05



User I/O

File: UserInputOutput.kicad_sch

Programming

Estimated BOM cost: \$0.00



Programming

File: Programming.kicad_sch

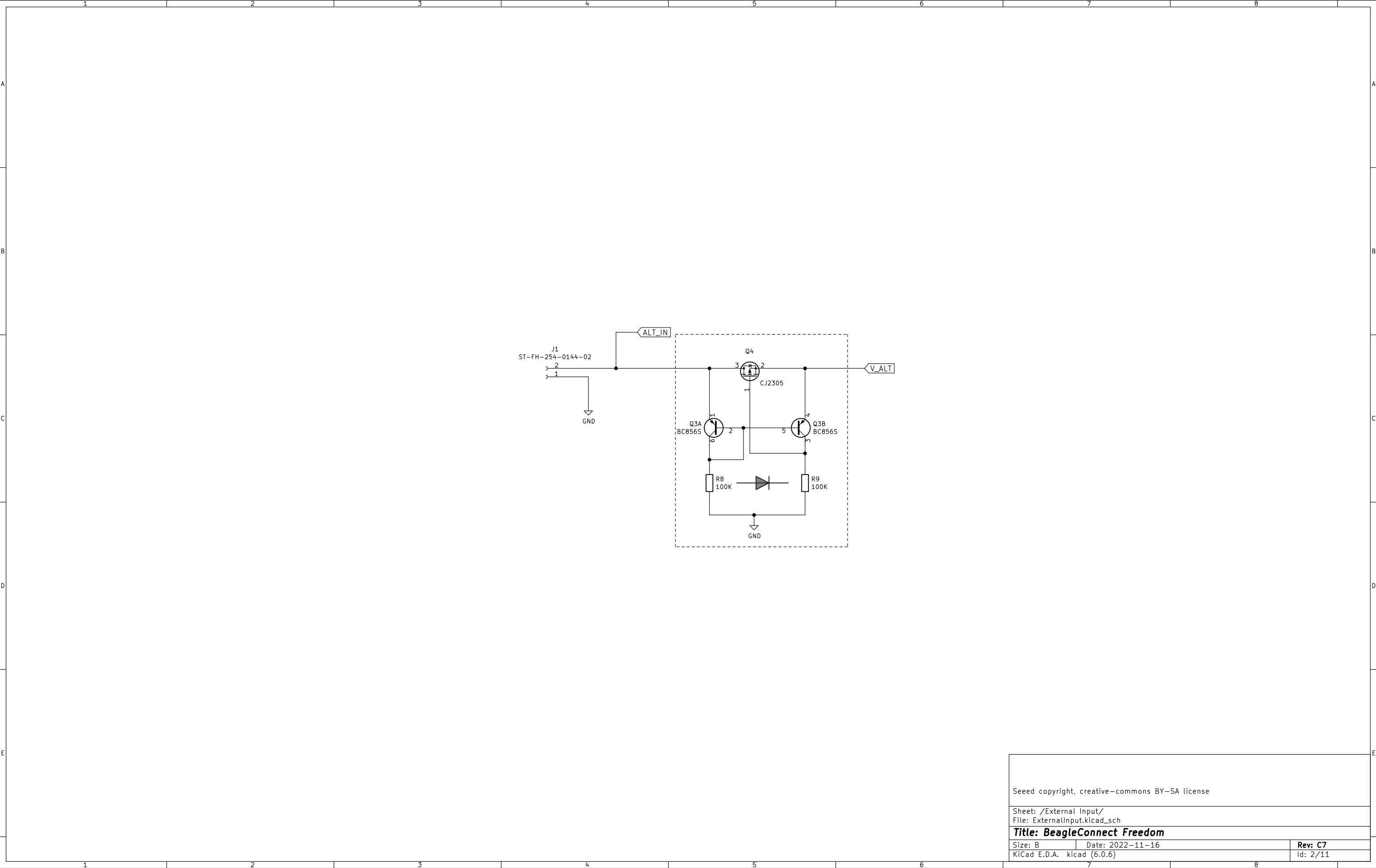
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Sheet: /
File: BeagleConnect Freedom C7.kicad_sch

Title: **BeagleConnect Freedom**

Size: C Date: 2022-11-16
KiCad E.D.A. kicad (6.0.6)

Rev: C7
Id: 1/11

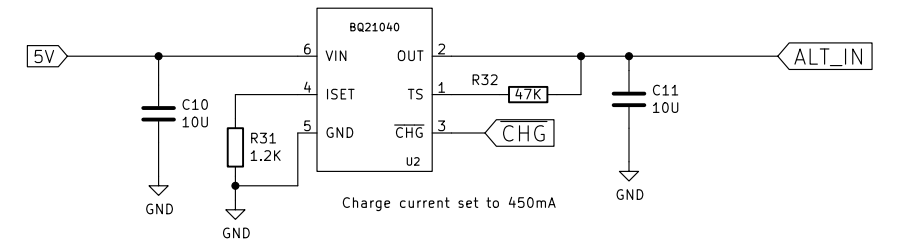


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Sheet: /External Input/
File: ExternalInput.kicad_sch

Title: BeagleConnect Freedom

Size: B	Date: 2022-11-16	Rev: C7
KiCad E.D.A. kicad (6.0.6)		Id: 2/11

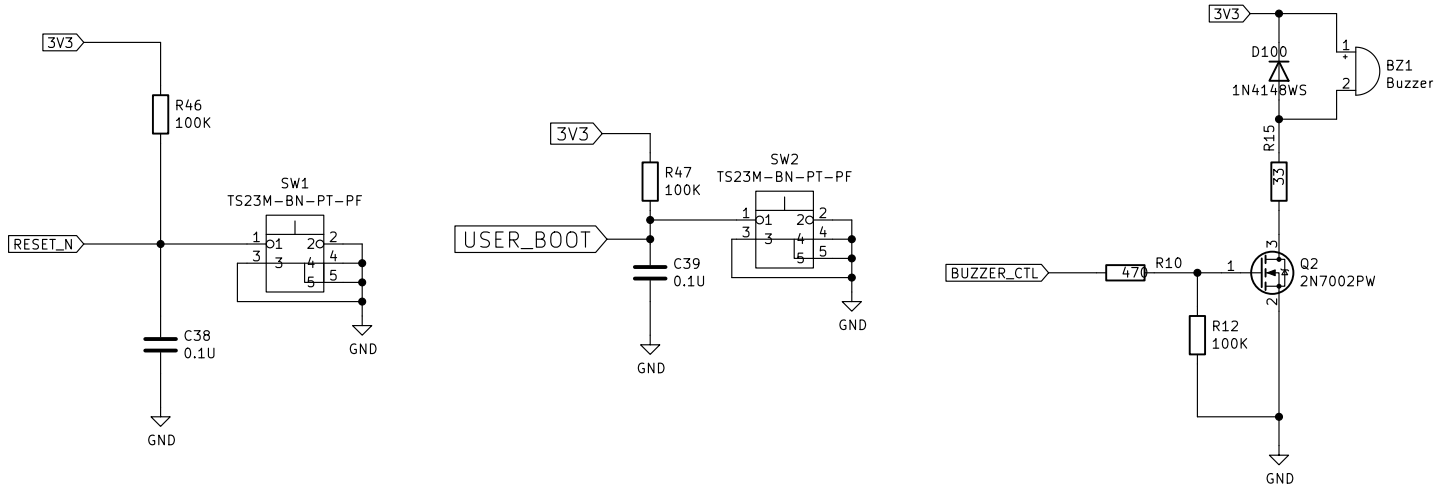


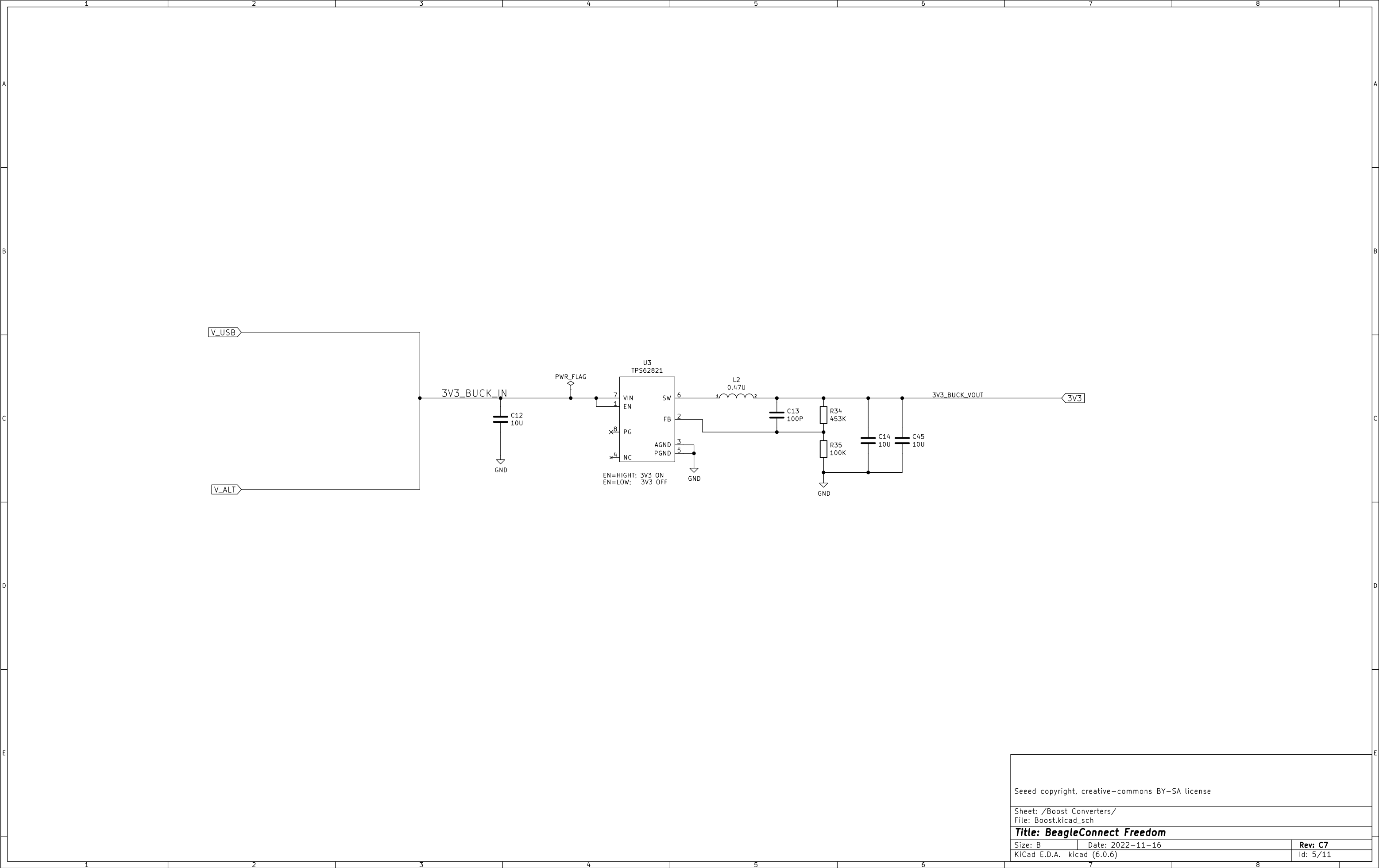
The USER_BOOT button is also used as an input to the CC1352R for it's programming, but is active low

On rev B prototype, the soldermask peeled back and many of the pins on the edge of the QFN would short to the center pad after attempted repair to the part (reflow, component replacement)

REV. C7

d: 3/11



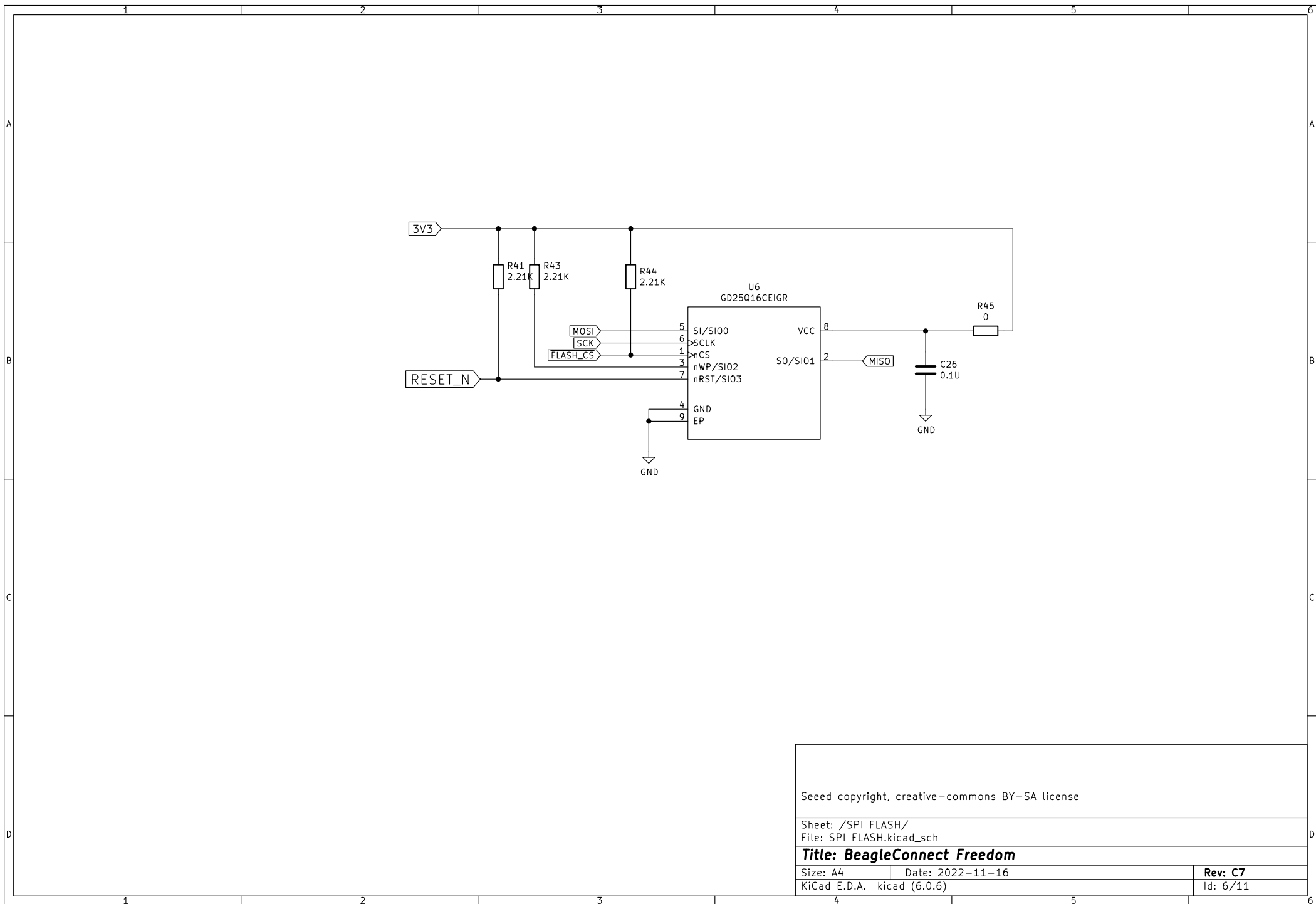


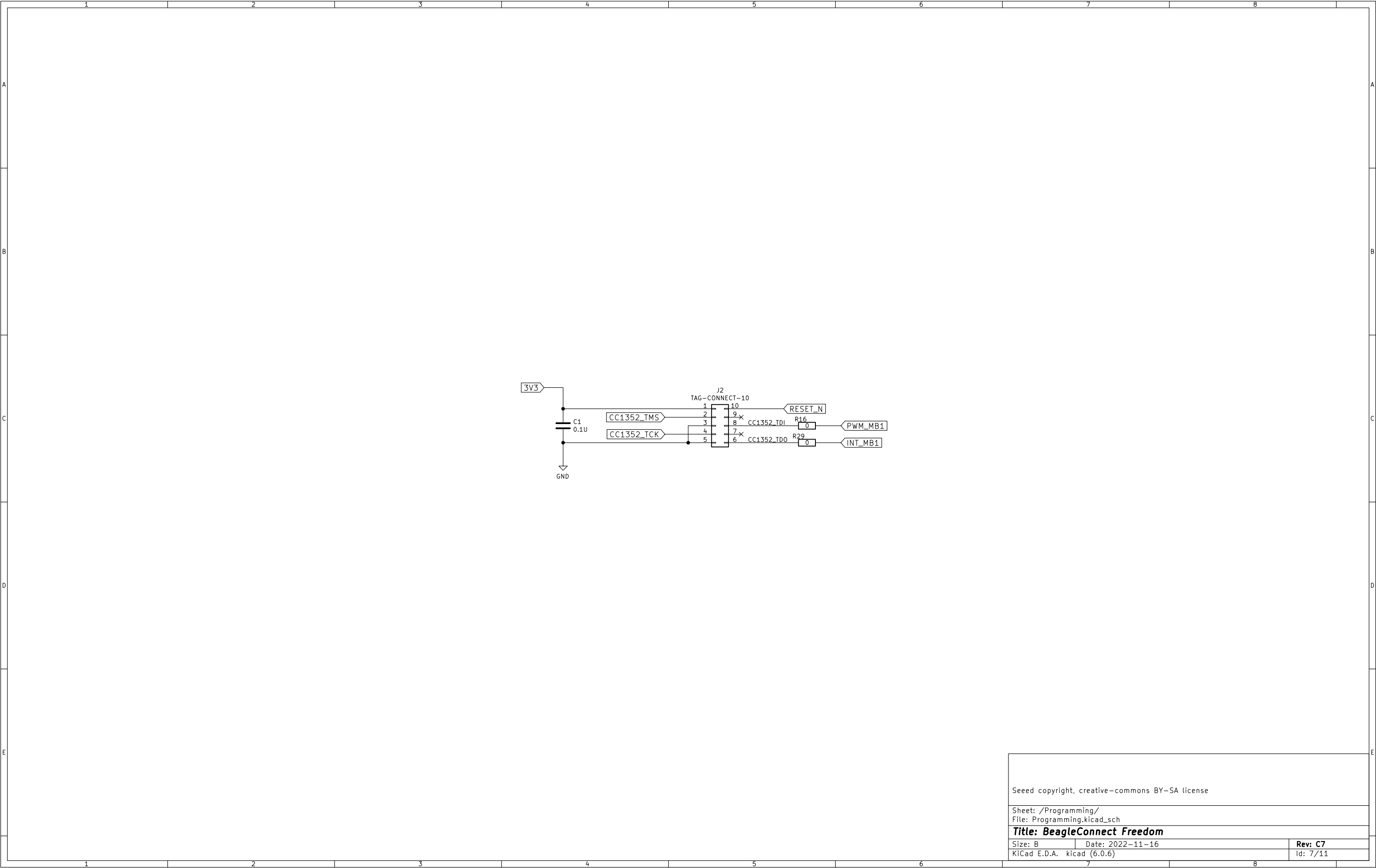
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Sheet: /Boost Converters/
File: Boost.kicad_sch

Title: BeagleConnect Freedom

Size: B	Date: 2022-11-16	Rev: C7
KiCad E.D.A. kicad (6.0.6)	Id: 5/11	



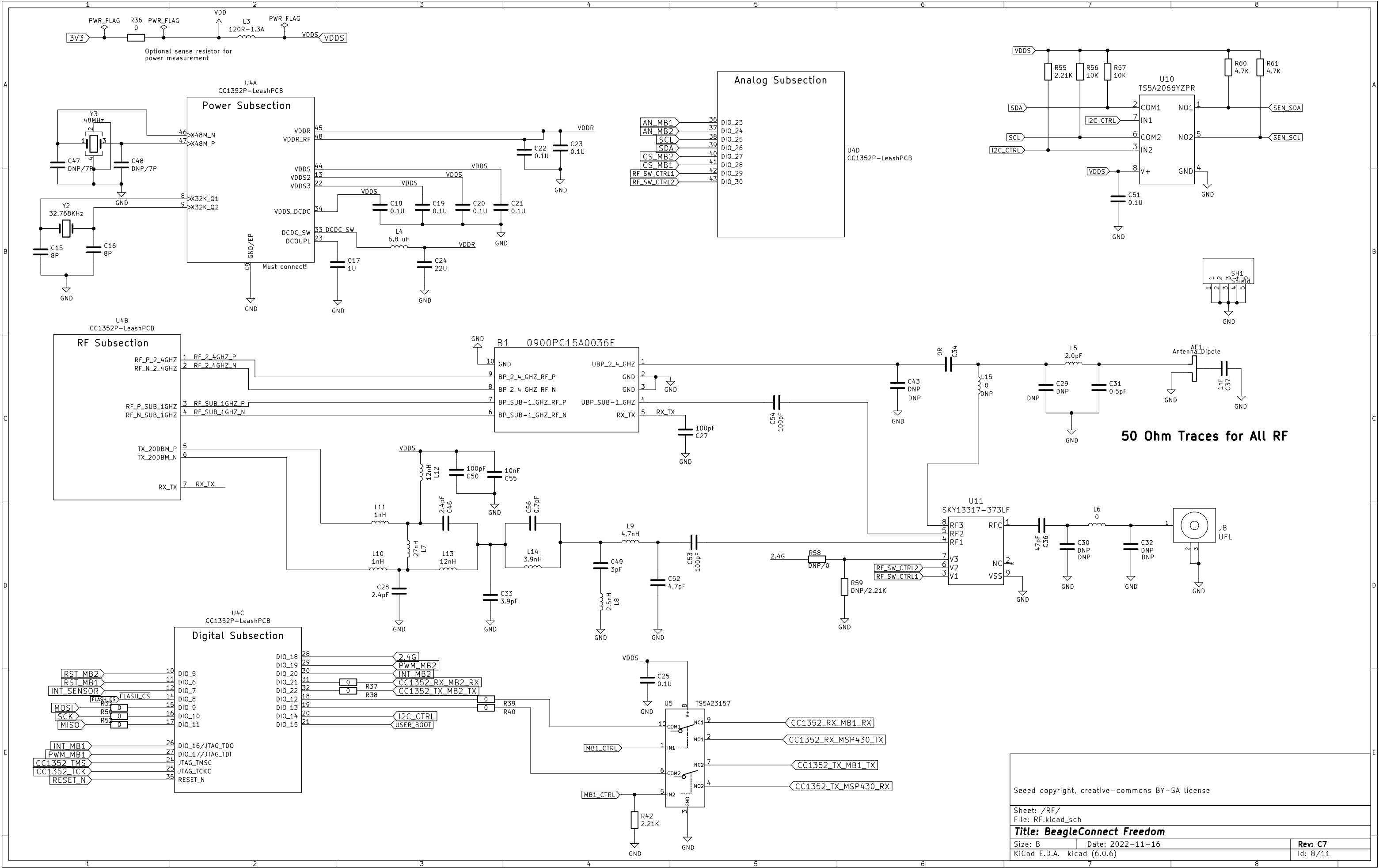


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Sheet: /Programming/
File: Programming.kicad_sch

Title: BeagleConnect Freedom

Size: B	Date: 2022-11-16	Rev: C7
KiCad E.D.A. kicad (6.0.6)		Id: 7/11

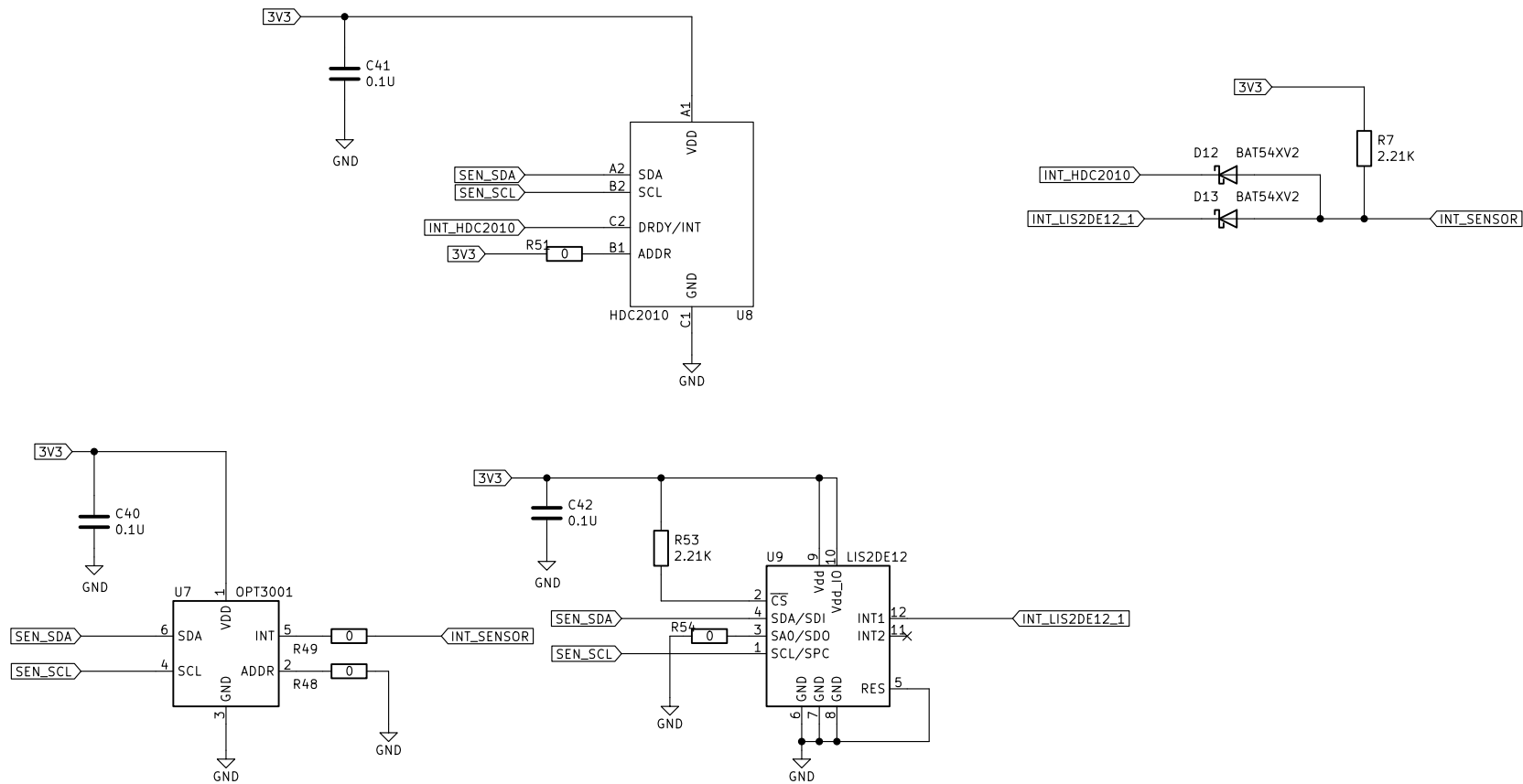


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Sheet: /RF/
File: RF.kicad_sch

Title: BeagleConnect Freedom

Size: B	Date: 2022-11-16	Rev: C7
KiCad E.D.A. kicad (6.0.6)		Id: 8/11



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Sheet: /Sensors/
File: Sensors.kicad_sch

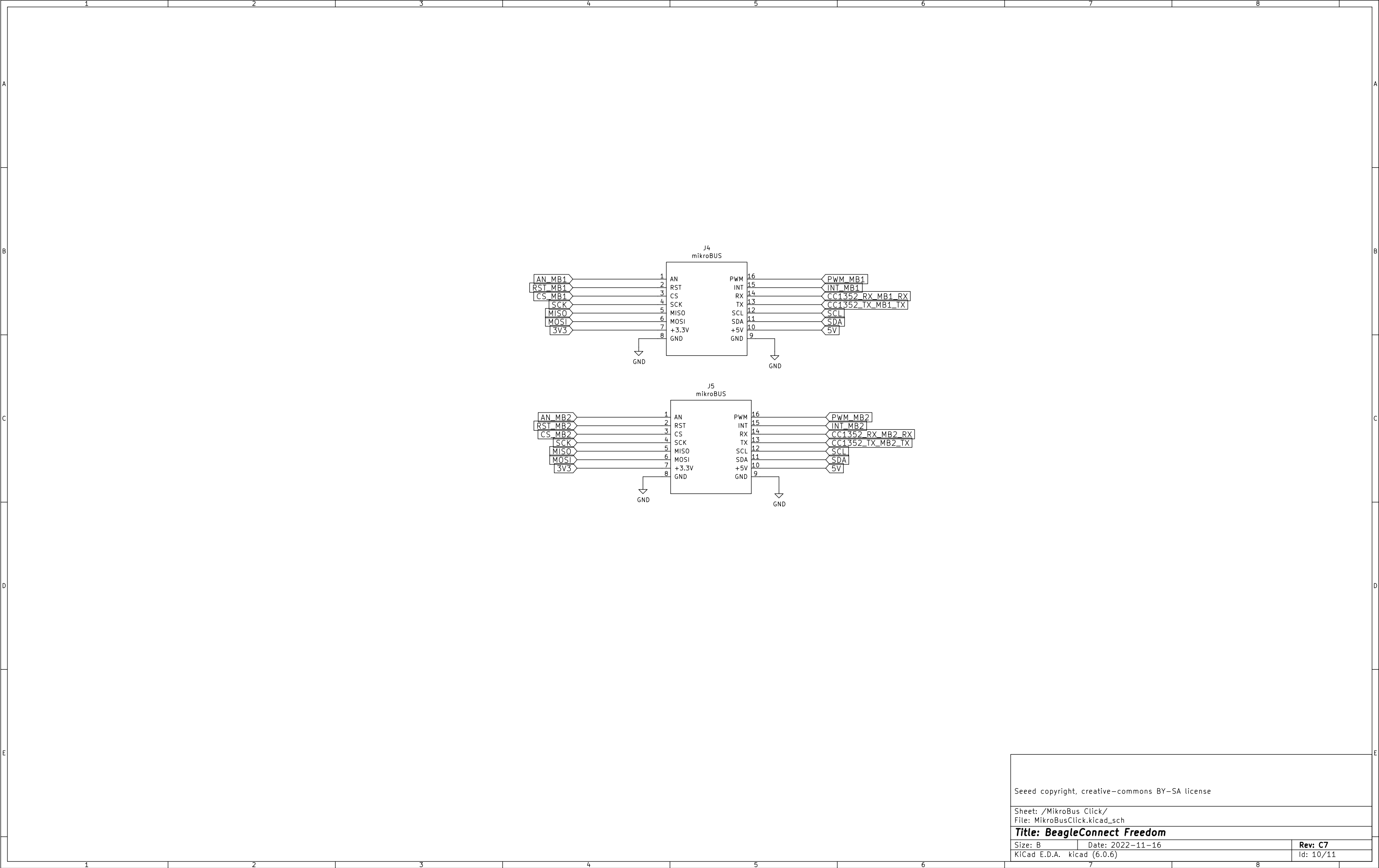
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Size: A4 Date: 2022-11-16

KiCad E.D.A. kicad (6.0.6)

Rev: C7

Id: 9/11

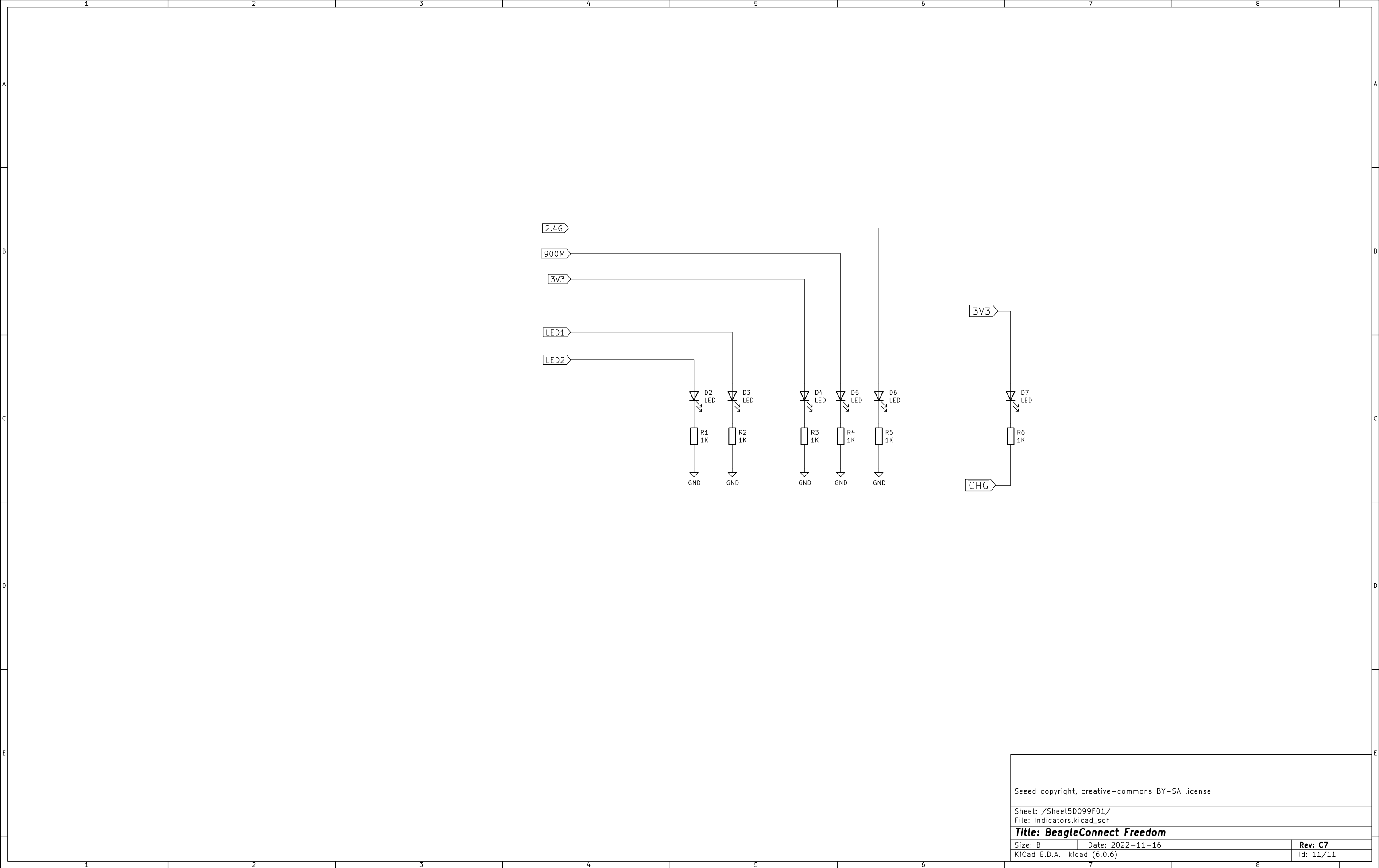


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Sheet: /MikroBus Click/
File: MikroBusClick.kicad_sch

Title: BeagleConnect Freedom

Size: B	Date: 2022-11-16	Rev: C7
KiCad E.D.A. kicad (6.0.6)	Id: 10/11	



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Sheet: /Sheet5D099F01/ File: Indicators.kicad_sch		
Title: BeagleConnect Freedom		
Size: B	Date: 2022-11-16	Rev: C7
KiCad E.D.A. kicad (6.0.6)		Id: 11/11